Module 7 · SW Design

Developer Testing

Tests are the safety net that lets you change code without fear.

Akın Kaldıroğlu · Lead Trainer

Why it matters

Tests Buy You Courage

Every principle so far assumed you can change code safely. Tests are what make that true — they turn 'I hope' into 'I know'.

Confidence

Refactor and extend without breaking what worked.

Feedback

Catch mistakes in seconds, not in production.

Design pressure

Hard-to-test code is usually badly designed.

The other half of defending correctness: defensive programming stops bad values at the barricade; tests prove the code behind it is actually right.

What you'll leave with

Learning Objectives

- Write clear unit tests with the Arrange–Act–Assert shape.
- Use test doubles to isolate the unit under test.
- Know when to reach for integration tests instead.
- See why testable code and good design are the same thing.

What tests catch

Bugs and Regressions

Two different failures — and your tests catch them at two different moments.

A bug

Behavior that never matched the requirement — wrong the first time anyone ran it. A new test catches it the first time it runs.

A regression

Behavior that used to be correct and broke — usually because of a change made somewhere else. An old test catches it every time it runs after.

regression — Latin regredi: re- (back) + gradi (to step). The code has literally stepped backwards.

A test earns its keep the second time it runs.

What tests catch

Turn Every Bug Into a Test

A bug report is a missing test. Reproduce it before you fix it — then keep the test forever.

Java

the off-by-one that shipped: orders of exactly 100 got no discount

```
// 1 · RED — reproduce the report as a failing test
@Test void appliesDiscountAtExactlyOneHundred() {
  assertEquals(90.0, order(100.0).total(), 0.001);   // fails: got 100.0
}

// 2 · GREEN — fix the boundary
-   return total >  100 ? 0.10 : 0.0;
+   return total >= 100 ? 0.10 : 0.0;

// 3 · KEEP — it stays in the suite; from now on it is a regression test
```

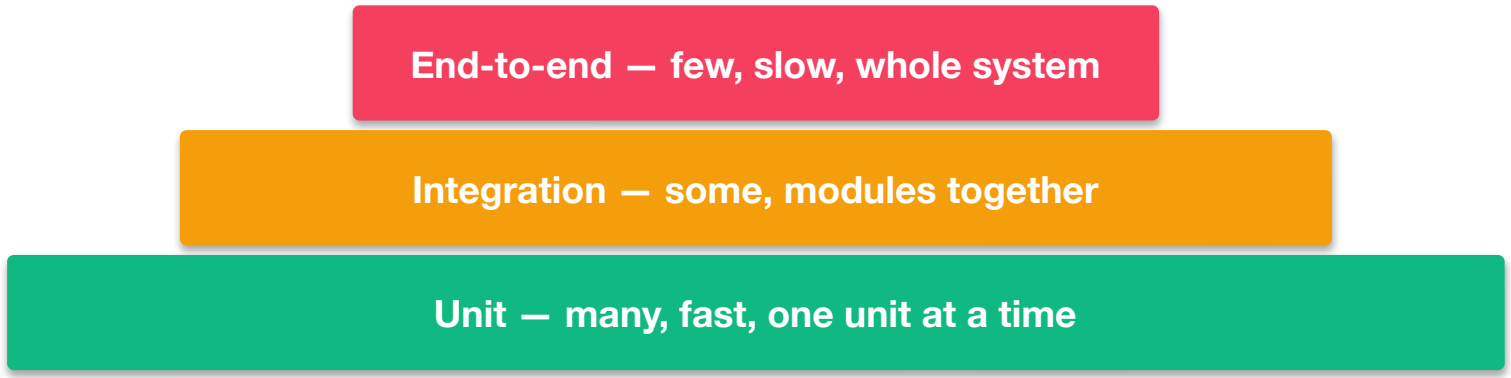
Never fix a bug without a failing test that proves it. Fix it without one and nothing stops it coming back.

For legacy code you don’t understand yet, characterization tests pin down what it does today (Feathers).

The strategy

The Test Pyramid

Many fast, focused tests at the bottom; a few slow, broad ones at the top.



Push tests down the pyramid: cheaper to write, faster to run, easier to trust.

Section 01

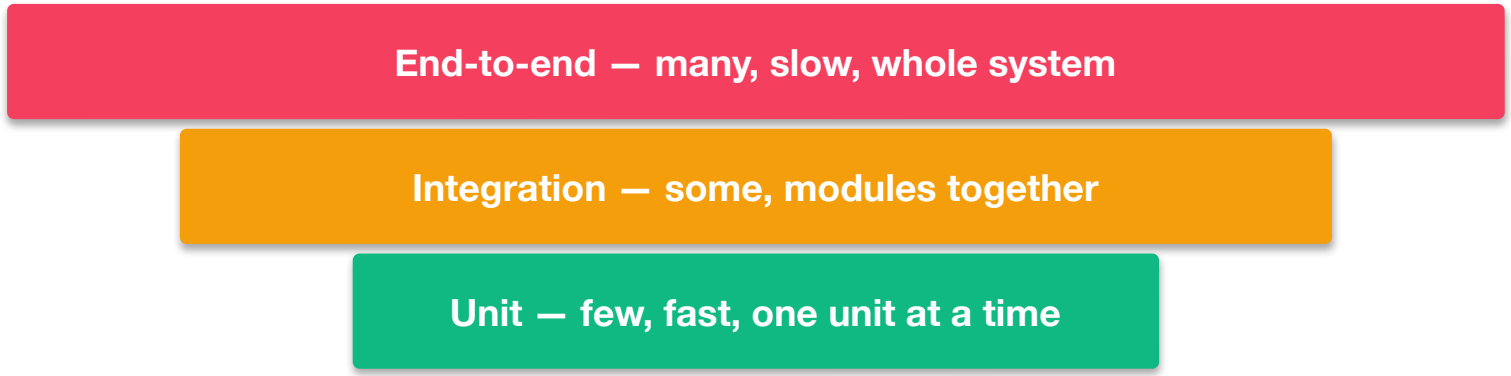
Unit Testing

Test one small unit in isolation — fast and focused.

The anti-pattern

The Ice-Cream Cone

In practice many teams end up here — the pyramid flipped upside down: many slow end-to-end tests, barely any unit tests.



- Slow feedback — a full run takes minutes, so it runs only in CI, hours later.
- Flaky — timing, network and shared data turn green red at random.
- Failures don’t localize — a red end-to-end test says ‘something broke’, not where.

The fix isn’t more end-to-end tests — it’s pushing behavior down into fast unit tests.

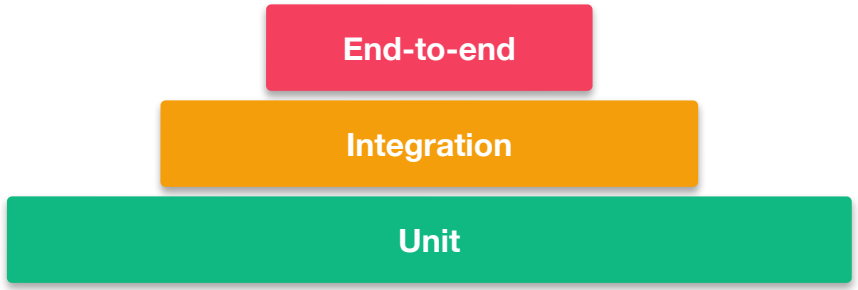
CI — continuous integration: *the build server that runs the whole suite automatically after every push, instead of you running it locally.*

The strategy

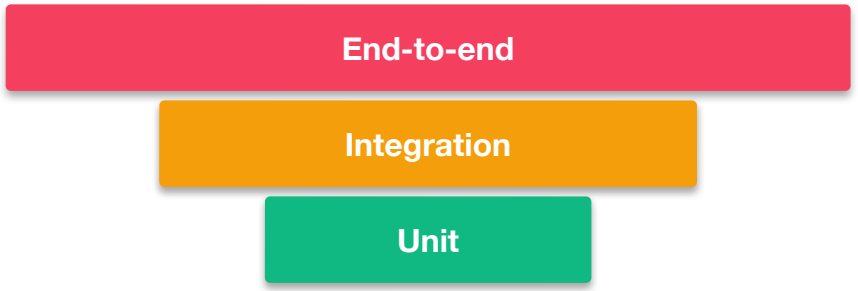
Two Shapes, Same Tests

The same test budget, spent two ways. The shape decides how fast you learn you broke something — and how precisely.

Pyramid — what you want



Ice-cream cone — what you get



	Pyramid	Ice-cream cone
Feedback	seconds, on your machine	minutes, in CI, hours later
A red test says	this behavior broke	something, somewhere broke
Flakiness	rare — no I/O to misbehave	routine — timing, network, data

The scope

What Is Developer Testing?

Testing you do while writing the code — not a phase handed to someone afterwards. It is unit testing plus integration testing, and the second has levels:

Unit

One behavior, in isolation, no I/O. Milliseconds. Most of your tests.

Integration

Your code plus something real — a module, a database, an endpoint. Seconds. Fewer.

Integration, from narrow to broad — each step trades speed for reach:

- **Narrow — a few classes together** *Still in-process, no I/O. Almost as fast as a unit test.*
- **Adapter — your code against one real dependency** *A repository against a real database; a client against a stub server.*
- **Contract — the agreement between two services** *Checked from both sides, so neither can break the other unnoticed.*
- **Broad — the whole application, wired up** *Driven through its own API. Slowest and least precise; keep very few.*

Beyond this sits end-to-end testing — the whole system, real externals, usually someone else's job. Developer testing stops where your code stops.

▶ runnable `dev.kaldiroglu.bootcamp.testing`

Arrange · Act · Assert

Every unit test tells a tiny story in three beats — the AAA pattern:

Arrange

build the object and its inputs

Act

call the one behavior under test

Assert

check the outcome — and nothing else

Java / JUnit

```
@Test
void addsTwoNumbers() {
  var calc = new Calc();   //
arrange
  int r = calc.add(2, 3);  // act
  assertEquals(5, r);      //
assert
}
```

C# / xUnit

```
[Fact]
public void AddsTwoNumbers() {
  var calc = new Calc();   //
arrange
  var r = calc.Add(2, 3);  // act
  Assert.Equal(5, r);      //
assert
}
```

Python / pytest

```
def test_adds_two_numbers():
  calc = Calc()            #
arrange
  r = calc.add(2, 3)       # act
  assert r == 5            #
assert
```

Unit

What Makes a Good Test Suite

A suite is more than a pile of tests. Three properties separate one that pays for itself from one that becomes a burden:

It is integrated into the development cycle

You only get value from tests you actively use. A suite that runs nightly — or only before a release — has already stopped paying you back.

It targets the parts that matter

Not all production code deserves equal attention. Cover the heart of the application — its domain model — before glue code, getters and configuration.

It gives maximum value for minimum maintenance

Every test is code you must keep working. A suite that costs more than it saves gets ignored first, then deleted.

> *“Program testing can be used to show the presence of bugs, but never to show their absence.”* — **Edsger W. Dijkstra**

Unit

Four Attributes of a Good Unit Test

A unit test exercises one unit with no database, network or file system. Judge every one of them against four attributes (Khorikov):

Protection against regressions

Does it actually catch a break? The more meaningful production code a test exercises, the more it protects.

Resistance to refactoring

Does it stay green when you improve the design without changing behavior? A test tied to implementation cries wolf.

Fast feedback

How soon does it tell you? Slow tests get run less often, and late feedback costs far more to act on.

Maintainability

How hard is it to read, and to keep running? An unreadable test is a future deletion.

The first three pull against each other — you can maximize any two, never all three. Maintainability is the one you never trade away.

Unit

What Is a ‘Unit’, Really?

‘One class’ is the easy answer. The useful one is a unit of behavior — a use case that may span a few classes. And ‘isolation’ means two different things:

	Classicist — Kent Beck	Mockist — London · GOOS
A ‘unit’ is	a behavior (may span classes)	usually a single class
‘Isolation’	tests isolated from each other	object isolated from collaborators
Collaborators	real; fake only I/O at the edges	mocked — verify the interactions
Mocks	sparing, at process boundaries	central — “mock roles, not objects”

Prefer the unit of behavior: tests that survive refactoring across classes and fail for one behavioral reason.

Unit

Isolation Is an Architecture Decision

A behavior is easy to unit-test exactly when it depends on nothing but itself. Whether that's true is decided by your architecture, not your test framework.

Rich domain · Hexagonal / DDD → unit = behavior

Rules live in dependency-free entities — call them and assert, no mocks. Mock only the ports at the edges (GOOS: "mock roles, not objects").

Anemic · a service over a repository → unit = class

Logic sits in a service that calls a repository, so you must mock the repo to test it — London by necessity, and brittle when you refactor.

Tangled · logic woven into SQL / HTTP → no unit at all

No seam to isolate — only integration and end-to-end tests fit. This is how a codebase arrives at the ice-cream cone.

Refactoring course — see which ACME Orders behaviors need doubles, and why → `github.com/kaldiroglu/refactoring-course-student-materials`

Unit

The Same Rule, Three Architectures

One business rule — an order total — written three ways in C#. The architecture, not the test framework, decides what a unit test can reach.

Rich domain · no doubles

```
// rule lives in the entity
public Money Total() =>
  lines.Aggregate(Money.Zero,
    (s,l) => s.Add(l.Subtotal()));

// test — nothing to mock
var o = new Order(Line(2, 10m));
Assert.Equal(20m, o.Total());
```

Anemic · mock the repo

```
// logic in a service over a repo
public decimal Total(int id) {
  var o = _repo.Find(id);
  return o.Lines.Sum(l => l.Total);
}
// test — the repo must be doubled
var repo = new Mock<IOrderRepo>();
repo.Setup(r => r.Find(1))
    .Returns(order);
```

Tangled · no seam

```
// rule buried in the query
var cmd = new SqlCommand(
  "SELECT SUM(qty*price) " +
  "FROM lines WHERE id=@id", conn);
var t = cmd.ExecuteScalar();

// no unit test reaches this —
// it needs a real database
```

No double · one mock · no seam at all. Same rule, same language — only the design changed. Testability is decided long before the first test is written.

Unit

Test Doubles Isolate the Unit

To test a unit alone, replace its collaborators with stand-ins. This is why we depend on abstractions (DIP) — so we can swap in a fake.

Common doubles

- Stub — returns canned answers
- Mock — verifies it was called right
- Fake — a lightweight working version

Why it works

- The unit depends on an interface…
- …so tests inject a double instead
- Design and testability go together

Unit

A Method Worth Testing

Read it by color — the rules and the outside calls are woven together:

- business rule
- reaches outside — each one forces a double
- plumbing

Java

every cyan call is a collaborator the test must stand in for

```java
public User changePassword(String username, PasswordChangeRequest change) {
  User currentUser = userRepository.findByUsername(username)
      .orElseThrow(() -> new UsernameNotFoundException("Username not found."));
  String currentPassword = currentUser.getPassword();
  String newPassword = encoder.encode(change.getNewPassword());
  TValidationException tex = new TValidationException("password");
  if (!isPasswordValid(change.getNewPassword()))
      tex.rejectNow("pattern", "not.valid");
  if (!encoder.matches(change.getOldPassword(), currentPassword))
      tex.rejectNow("old", "not.matches");
  if (encoder.matches(change.getNewPassword(), change.getOldPassword()))
      tex.rejectNow("old", "same");
  currentUser.setPassword(newPassword);
  return updateUser(currentUser);
}
```

Amber and cyan interleave — that is the anemic tangle: you cannot reach a rule without going through a port.

Unit

Unit Test With Mocks

mock setup act assert plumbing

Java · JUnit 5 + Mockito

eight cyan lines of setup — for one act and two asserts

```java
@Mock UserRepository userRepository;
@Mock PasswordEncoder encoder;
@InjectMocks UserService service;
@Test void changesPasswordWhenRequestIsValid() {
  User user = new User("ada", "ENC(old)");   // arrange
  var change = new PasswordChangeRequest("oldPass", "newPass1!");
  when(userRepository.findByUsername("ada")).thenReturn(Optional.of(user));
  when(encoder.encode("newPass1!")).thenReturn("ENC(new)");
  when(encoder.matches("oldPass", "ENC(old)")).thenReturn(true);
  when(encoder.matches("newPass1!", "oldPass")).thenReturn(false);
  when(userRepository.save(user)).thenReturn(user);
  User result = service.changePassword("ada", change);   // act
  assertThat(result.getPassword()).isEqualTo("ENC(new)");  // assert
  verify(userRepository).save(user);
}
```

Classify it: anemic — the middle column of page 16. Two mocks are needed because the rules live in the service, not in the model.

Unit

Move the Rules, Lose the Mocks

The same feature, with the rules moved into the domain:

rules in the domain

```
class User {
  void changePassword(Password current,
                      Password next,
                      PasswordHasher hasher) {
    if (!hasher.matches(current, hash))
      throw new OldPasswordMismatch();
    if (current.equals(next))
      throw new SamePassword();
    hash = hasher.hash(next);
  }
}
// Password.of(raw) already rejected an
// invalid pattern — see Deck 02
```

its test — one fake, no mocks

```
@Test void rejectsWhenOldDoesNotMatch() {
  var user = userWith("ENC(old)");
  var hasher = new FakeHasher();
  assertThrows(OldPasswordMismatch.class,
    () -> user.changePassword(
      Password.of("wrong1!"),
      Password.of("newPass1!"), hasher));
}

// no @Mock · no when(...) · no verify(...)
// the rule is tested where it lives
```

Two mocks became one fake — and the rejection path, where the real bug hides, is finally easy to test.

Section 02

Integration Testing

Do the parts actually work together?

Integration

Test the Seams

Units can each pass while the wiring between them fails. Integration tests exercise real collaborators — a real database, a real HTTP call.

- Catch mismatches units can't: SQL, serialization, configuration.
- Slower and fewer — use them for the risky seams, not everything.
- Often run against a real DB in a container for fidelity.

Integration

An Integration Test in C#

No doubles anywhere: the real routing, the real DI container, the real database — the whole pipeline, exercised from the outside.

act assert

C# · xUnit + WebApplicationFactory *one call exercises routing → model binding → DI → service → EF Core → SQL*

```
public class OrdersApiTests(WebApplicationFactory<Program> f)
    : IClassFixture<WebApplicationFactory<Program>> {
  private readonly HttpClient _client = f.CreateClient();   // the real app, in memory
  [Fact]
  public async Task PostOrder_CreatesItAndReturns201() {
    var response = await _client.PostAsJsonAsync("/orders",
        new { qty = 2, unitPrice = 10m });
    Assert.Equal(HttpStatusCode.Created, response.StatusCode);
    var created = await response.Content.ReadFromJsonAsync<OrderDto>();
    Assert.Equal(20m, created!.Total);
  }
}
```

Catches what no unit test can: routing, model binding, DI wiring, the JSON contract, SQL and transactions.

Pays for it: seconds instead of milliseconds, a database to keep alive, and a failure that says “somewhere in that chain”.

Integration

Unit vs Integration

	Unit	Integration
Scope	One class / function	Several parts together
Collaborators	Replaced with doubles	Real (DB, queue, API)
Speed	Milliseconds	Seconds
Count	Many	Some
Catches	Logic errors	Wiring & contract errors

A technique

Red · Green · Refactor

Test-Driven Development flips the order: write a failing test first, then the code to pass it, then clean up. Tests become your design tool.

Red

Write a small failing test for the next behavior.

Green

Write the simplest code that makes it pass.

Refactor

Improve the design — tests keep you safe.

A technique

The Discipline: Three Laws

Kent Beck's loop, tightened into three rules by Robert C. Martin. They exist to keep each step small enough that you never lose control:

1. **Write no production code until a failing test demands it.**
   The test comes first — always. Otherwise you are writing code nothing asked for.

2. **Write no more of a test than is enough to fail.**
   Not compiling counts as failing. Stop the moment it is red and go make it pass.

3. **Write no more production code than is enough to pass that test.**
   The simplest thing that works. Generality waits until a test asks for it.

Each turn of the loop is seconds long. If a step takes ten minutes, the step was too big.

The by-product: you end up with only the code some test asked for — YAGNI, enforced by the loop rather than by willpower.

A technique

Test First or Test After?

The loop is the same either way. The order changes what the test is for — and whether you can trust it.

Test after — code, then test

- Born green — you never watch it fail
- Describes the code, not the requirement
- Untestable design found too late
- “Later” tests that never arrive

Test first — test, then code

- You watch it fail, so you know it can
- States the requirement, not the solution
- You are the first caller of your own API
- Done means green — scope stops growing

A test that has never failed is not a safety net — it is a decoration.

Not a law: characterization tests for legacy code are written after on purpose (Feathers), and a spike can be thrown away and redone test-first.

Zoom out

Testable Means Well-Designed

If a unit is hard to test, that's a design smell — usually hidden dependencies or too many responsibilities.

Hard to test →

- Tight coupling to concretes
- Doing too many things (low cohesion)
- Hidden state and side effects

Easy to test →

- Depends on injected abstractions
- One responsibility, clear inputs/outputs
- Pure where possible

The AI angle

AI Writes Tests — You Own Coverage

AI is excellent at generating test scaffolding and edge cases. But a test is only as good as the behavior it pins down.

- Ask for tests of edge cases: empty, null, boundary, error paths.
- Check that tests assert behavior — not just that code ran.
- Beware tests written to match buggy code; you decide what 'correct' is.

Remember these

Key Takeaways

- ✓ Tests give you the courage to change code safely.
- ✓ Unit tests: one unit, isolated, fast — Arrange, Act, Assert.
- ✓ Test doubles isolate the unit; they rely on depending on abstractions.
- ✓ Integration tests check the seams with real collaborators.
- ✓ Hard to test usually means badly designed — testability is a design goal.

Go deeper

Further Reading

The books behind this session. If you read only one, read the first.

📖 **Test Driven Development: By Example** Kent Beck · Addison-Wesley, 2002

The red-green-refactor loop from its source, worked end to end.

📖 **Growing Object-Oriented Software, Guided by Tests** Freeman & Pryce · Addison-Wesley, 2009

How tests drive design — where mocks belong and what they are really for.

📖 **Unit Testing: Principles, Practices, and Patterns** Vladimir Khorikov · Manning, 2020

The modern synthesis: what makes a test valuable, and why most suites are brittle.

📖 **Working Effectively with Legacy Code** Michael Feathers · Prentice Hall, 2004

Seams and characterization tests — how to test code that was not built for testing.

Fowler — “Mocks Aren’t Stubs”, the test-double taxonomy → `martinfowler.com/articles/mocksArentStubs.html`

Next session

What's Next

Module 8 · Saturday 1 August

Agentic Design & Coding

with Akın Kaldıroğlu

Putting it all together: how to design and build software with an AI agent — while you stay the decision-maker.

Further practice

Put a Safety Net Under It

Before you change the ACME Orders code, pin its behavior with characterization tests — then refactor without fear.

Refactoring course · Characterization Tests

Repo `github.com/kaldiroglu/refactoring-course-student-materials`

IN THIS SESSION, LOOK AT

- `Exercises/Ex04-CharacterizationTests.md`
- `Slides-PDF/Ch06-SafetyNet.pdf`
- `.../src/test/...` (the ACME Orders test suite)

Questions?

Test to move fast — safely.

Akın Kaldıroğlu · Developer Testing